



Paper id: 252272

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Subject Code: BMI201

Roll No:

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MCAINT
(SEM II) THEORY EXAMINATION 2024-25
DIGITAL ELECTRONICS

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****02 x 7 = 14**

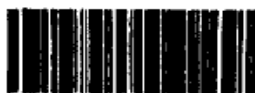
Q no.	Question	CO	Level
a.	Define digital system. What is the key difference between a digital and an analog system?	1	K2
b.	What are canonical forms? Differentiate between SOP and POS form.	2	K2
c.	Define Prime Implicant and Essential Prime Implicant in Boolean simplification.	3	K2
d.	Use a 2-variable K-map to simplify the Boolean function $F(A,B)=\sum m(1,2)$	3	K3
e.	Differentiate between multiplexer and demultiplexer.	4	K2
f.	What is a combinational circuit? How is it different from a sequential circuit?	4	K2
g.	What is a shift register? Mention one application each of SIPO and PISO registers.	5	K3

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

Q no.	Question	CO	Level
a.	Explain the differences between analog and digital systems. Describe the basic structure of a digital computer and explain how binary numbers are used in its operation.	1	K2
b.	State and prove the following basic theorems of Boolean algebra using truth tables: - Commutative Law - Associative Law - Distributive Law - Identity Law	2	K3
c.	Simplify the Boolean function using a 4-variable Karnaugh Map and implement the simplified expression using only NAND gates: $F(A,B,C,D)=\sum m(1,3,7,11,15)+d(0,2,5,13)$	3	K3
d.	What is a multiplexer? Draw the logic diagram of 8 to 1 line multiplexer.	4	K2
e.	Explain the different types of flip-flops along with their truth tables and characteristic equations	5	K2

SECTION C**3. Attempt any one part of the following:****07 x 1 = 07**

Q no.	Question	CO	Level
a.	Convert the following binary numbers to their 1's and 2's complements: (a) 10101011 (b) 00011100 (c) 11100001 Also explain how negative numbers are represented in signed 2's complement format.	1	K3



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b.	Explain different types of binary codes with examples: • BCD (Binary Coded Decimal) • Gray Code • Excess-3 Code • ASCII	1	K2
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4. Attempt any one part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Explain the difference between canonical and standard forms of Boolean expressions. Use examples to show how a standard SOP can be converted into a canonical SOP.	2	K2
b.	Define the following basic logic gates with symbols, truth tables, and Boolean expressions: • AND • OR • NOT	2	K2

5. Attempt any one part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Using the k-Map, simplify the Boolean function: $F(A,B,C,D) = \sum m(0,1,2,4,5,8,10)$	3	K3
b.	Find the canonical form of POS from the canonical form of given SOP. $F(A,B,C,D) = \sum m(0,2,4,6)$ https://www.aktuonline.com	3	K3

6. Attempt any one part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Design a full adder using NAND Gates.	4	K2
b.	Design a code converter that converts BCD to gray code.	4	K3

7. Attempt any one part of the following:		07 x 1 = 07	
Q no.	Question	CO	Level
a.	Explain the operation of universal shift register.	5	
b.	Define counter? Write the classification of counters?	5	